

1 Combining the existing spectral and overlap controls on the original source

This attachment records the exact comparison already established in the coordinating source intake. The complete earlier AW proof is retained in this reader; SP reproduces its spectra and final condition-number bound and adds the stated overlap refinement. Neither bound is discarded.

Fix the actual full packet h , tensor order $k \geq 1$, and its original cyclic polynomial χ of degree $q \geq 1$. The common coefficient space is $H = \mathcal{P}_{2q}$. Its integration coordinate is $S = k/2 + iy$. Write $x \in [0, 1]$ for metric interpolation and reserve u for the geometric deformation parameter of BC. The exact AW/SP coordinate map is

$$y = u_{AW}, \quad x = t_{AW}, \quad H_{AW} \xrightarrow{\text{id}} H_{SP}. \quad (\text{ACM1})$$

This identity leaves every polynomial coefficient unchanged. The arithmetic density is w_h^{*k} in both calculations. For the reference density, the original constant $c_a = 2^{1-2a}\Gamma(2a)$ gives $c_{1/4} = \sqrt{2\pi}$, so the exact convolution formula is

$$r_{1/4}^{*k}(y) = \frac{c_{1/4}^k}{c_{k/4}} r_{k/4}(y) = \frac{(\sqrt{2\pi})^k}{c_{k/4}} \frac{|\Gamma(k/4 + iy/2)|^2}{2\pi}. \quad (\text{ACM2})$$

Consequently the densities, their masses and all monomial integrals coincide. In particular $M_{AW}(x) = M_{SP}(x) = M_x$ on H , and $M_{N,AW} = I_N^* M_x I_N$ on the original subspace $I_N : \mathcal{P}_N \hookrightarrow H$.

AW's relation map has codomain $\mathcal{P}_N$, whereas SP incorporates its inclusion into H . Their literal equality is

$$B_N^{SP} = I_N B_N^{AW}. \quad (\text{ACM3})$$

Substitution into the orthogonal projector formula gives

$$\begin{aligned} P_N &= I_N (I_N^* M_x I_N)^{-1} I_N^* M_x, \\ Q_N &= I_N B_N^{AW} ((B_N^{AW})^* I_N^* M_x I_N B_N^{AW})^{-1} (B_N^{AW})^* I_N^* M_x. \end{aligned} \quad (\text{ACM4})$$

The two definitions of each projector therefore agree, including $Q_{q-1} = 0$ for the zero relation domain. Put

$$\begin{aligned} U_x &= Q_{2q-1} + Q_{2q} - Q_q = \mathcal{R}_x^{SP}, \\ W_x &= (P_{2q-1} - P_{q-1}) + (P_{2q} - P_q) = \mathcal{P}_x^{SP}, \\ C_x &= M_x^{-1} \dot{M}_x = T_x^{SP}. \end{aligned} \quad (\text{ACM5})$$

These are AW's original U, W, C . Its independently constructed source isometry T_x^{iso} satisfies

$$(T_x^{\text{iso}})^* M_x T_x^{\text{iso}} = M_x, \quad U_x = T_x^{\text{iso}} W_x (T_x^{\text{iso}})^{-1}. \quad (\text{ACM6})$$

Thus cyclicity of the same finite trace proves

$$\text{Tr}((U_x - W_x)C_x) = \text{Tr}(W_x (T_x^{\text{iso}})^{-1} [C_x, T_x^{\text{iso}}]). \quad (\text{ACM7})$$

The original observation is still $\mathcal{O} = \sigma_h^{\otimes k} \eta J_{2q}$. Its exact difference and kernel after the isometry remain

$$\begin{aligned} \mathcal{O} T_x^{\text{iso}} - \mathcal{O} &= \sigma_h^{\otimes k} \eta J_{2q} (T_x^{\text{iso}} - I), \\ \ker(\mathcal{O} T_x^{\text{iso}}) &= (T_x^{\text{iso}})^{-1} (\chi \mathcal{P}_q). \end{aligned} \quad (\text{ACM8})$$

The first equality is expansion. The second follows because $\sigma_h^{\otimes k}$ and η are the original injections and $\ker J_{2q} = \chi \mathcal{P}_q$. This retains the actual cohomology defect of the isometry rather than assigning it an intertwining property.

Let $F(x)$ be the original logarithmic four-endpoint ratio and $\Delta_{h,k} = F(1) - F(0)$. Its signed identity is $F'(x) = \text{Tr}((U_x - W_x)C_x)$ by (AW6), or equivalently (SP4)–(SP5) and (ACM5). Both U_x, W_x have the complete spectrum 0 with multiplicity q , 1 with multiplicity 2, and 2 with multiplicity $q - 1$. All statements below also cover $q = 1$.

Write $c_1(x) \leq \dots \leq c_{2q+1}(x)$ for the eigenvalues of C_x in the original M_x metric and $d_x = c_{2q+1}(x) - c_1(x)$. Let $s_x = \dim(\text{ran } U_x \cap \text{ran } W_x)$. The two already proved pointwise bounds, on this identical source, are

$$\begin{aligned} S_{\text{AW}}(x) &= c_{q+2}(x) - c_q(x) + 2 \sum_{j=1}^{q-1} (c_{2q+2-j}(x) - c_j(x)), \\ S_{\text{SP}}(x) &= \frac{d_x}{2} \min \left\{ 4q - 2s_x, \sqrt{(2q+1)(8q-4-2\text{Tr}(U_x W_x))} \right\}. \end{aligned} \quad (\text{ACM9})$$

For completeness, their simultaneous application follows directly as follows. An orthogonal projection P of rank r satisfies $\sum_{j=1}^r c_j \leq \text{Tr}(PC_x) \leq \sum_{j=2q+2-r}^{2q+1} c_j$: in an orthonormal eigenbasis of C_x , its diagonal entries lie in $[0, 1]$ and have sum r , so moving their mass to the lowest or highest eigenvalues proves each inequality. Each of U_x, W_x is the sum of its range projection of rank $q + 1$ and its eigenvalue-two projection of rank $q - 1$. Apply these bounds to both sums and subtract their extrema to obtain $|F'(x)| \leq S_{\text{AW}}(x)$, with every multiplicity retained.

The range dimensions give $s_x \geq 1$. If E_x projects onto the intersection, then $U_x - E_x$ and $W_x - E_x$ are positive, each of trace $2q - s_x$. Therefore $\|U_x - W_x\|_1 \leq 4q - 2s_x$. The same complete spectra give

$$\text{Tr}(U_x - W_x) = 0, \quad \text{Tr}((U_x - W_x)^2) = 8q - 4 - 2\text{Tr}(U_x W_x). \quad (\text{ACM10})$$

Cauchy–Schwarz on all $2q + 1$ real eigenvalues gives the other trace-norm bound in (ACM9). Subtract the midpoint $(c_1 + c_{2q+1})I/2$ from C_x in the trace pairing; the first identity in (ACM10) leaves F' unchanged. Diagonalizing $U_x - W_x$ then bounds the pairing by $d_x \|U_x - W_x\|_1/2$. This proves $|F'(x)| \leq S_{\text{SP}}(x)$ in the same original metric.

Every subsequent bound is now evaluated on this identical path, including the centered Hilbert–Schmidt, angle and signed finite-residual controls. For compact notation put $n = 2q + 1$, $a_q = 2q - 1$, and define

$$\begin{aligned} T_q(x) &= \frac{1}{2} d_x \|U_x - W_x\|_{1, M_x}, \\ S_q(x) &= \sqrt{\left(\text{Tr}(C_x^2) - \frac{(\text{Tr } C_x)^2}{n} \right) \text{Tr}((U_x - W_x)^2)}, \\ z_x &= a_q \sqrt{1 - e^{-F(x)/a_q}}, \quad A_q(x) = z_x d_x, \\ J_q &= \int_0^1 \min\{S_{\text{AW}}(x), T_q(x), S_{\text{SP}}(x), S_q(x), A_q(x)\} dx, \\ I_q &= \int_0^1 \min\{d_x, S_{\text{AW}}(x)/z_x, T_q(x)/z_x, S_{\text{SP}}(x)/z_x, S_q(x)/z_x\} dx. \end{aligned} \quad (\text{ACM11a})$$

These symbols are exactly $E_q, T_q, O_q, S_q, A_q, J_q, I_q$ of (RMT9)–(RMT13), with $E_q = S_{\text{AW}}$ and $O_q = S_{\text{SP}}$. Equations (ACM1)–(ACM5) prove the operator and measure identities needed for that substitution. In particular they retain the actual inverse metric in the signed density $v(U_x - W_x)M_x^{-1}v^*$.

Here is the proof of the additional terms at this earlier use site. Trace zero in (ACM10) allows subtraction of $\text{Tr } C_x/n$ from C_x ; Hilbert–Schmidt Cauchy–Schwarz gives $|F'| \leq S_q$. The midpoint argument already proved gives $|F'| \leq T_q \leq S_{\text{SP}}$. Retain both complete q -entry angle lists of the minimum-section pairs $(q - 1, 2q)$ and $(q, 2q - 1)$. The second pair has a designated zero angle by the original ambient dimension. Each additional zero remains in its list. The exact projection-difference spectra give $|F'| \leq d_x \sum \sin \theta$ over the two lists. Their cosine-square determinants give $F = \sum -\log \cos^2 \theta$; applying concavity of $t \mapsto \sqrt{1 - e^{-t}}$ to the remaining $2q - 1$ entries gives

$|F'| \leq A_q$, exactly as proved with every multiplicity in (RMT11). The designated zero still contributes its displayed zero to both scalar sums.

The denominator in (ACM11a) is strictly positive on this actual path. Indeed equality $F = 0$ makes both crossed minimum sections equal: their Gram difference is the squared norm of their difference, and every relative eigenvalue is at least one. Equality of their determinants makes every such eigenvalue one. In particular the constant remainder 1 would be orthogonal to $\chi\mathcal{P}_q$. The polynomial $\chi^{\#k}(S) = \overline{\chi(k - \bar{S})}$ has degree q and equals $\overline{\chi(S)}$ on $S = k/2 + iy$. That orthogonality would imply $\int |\chi|^2 d\mu_x = 0$, contradicting the positive Gram of the nonzero polynomial χ . This proves $F(x) > 0$ without inserting a new hypothesis. Compactness gives a positive minimum. All remaining functions are continuous except the intersection dimension, whose rank strata are Borel sets defined by matrix minors. Thus every displayed integral is finite. Integrating the five bounds on the same signed integrand proves

$$|\Delta_{h,k}| \leq J_q \leq \int_0^1 \min\{S_{AW}(x), S_{SP}(x)\} dx. \quad (\text{ACM11})$$

The earlier minimum remains an explicit majorant of the current bound. If $b_1 \leq \dots \leq b_n$ are the positive eigenvalues of $D = M_0^{-1}M_1$ in M_0 , direct inversion of $M_x = M_0((1-x)I + xD)$ gives

$$c_j(x) = \frac{b_j - 1}{1 - x + xb_j}, \quad \int_0^1 c_j(x) dx = \log b_j. \quad (\text{ACM12})$$

The order is preserved because the derivative with respect to b_j is $(1 - x + xb_j)^{-2} > 0$. Termwise integration of the finite sum gives

$$|\Delta_{h,k}| \leq J_q \leq \min \left\{ \mathcal{L}_q(D), \int_0^1 T_q dx, \int_0^1 S_{SP} dx, \int_0^1 S_q dx, \int_0^1 A_q dx \right\}, \quad (\text{ACM13})$$

$$\mathcal{L}_q(D) = \log \frac{b_{q+2}}{b_q} + 2 \sum_{j=1}^{q-1} \log \frac{b_{2q+2-j}}{b_j} \leq (2q-1) \log \frac{b_n}{b_1}.$$

The $q = 1$ sum is empty. The last inequality follows by bounding each of its $1 + 2(q-1)$ logarithmic factors by $\log(b_n/b_1)$.

The nonlinear coordinate also receives every one of these controls. For $B > 0$ put

$$\mathcal{H}_{a_q}(B) = 2 \operatorname{arcosh}(e^{B/(2a_q)}), \quad \Psi_{a_q}(t) = 2a_q \log \cosh(t/2) \quad (t \geq 0). \quad (\text{ACM14})$$

Their two inverse composites on the positive half-lines are identities. Direct differentiation gives

$$\mathcal{H}'_{a_q}(B) = \frac{1}{a_q \sqrt{1 - e^{-B/a_q}}}.$$

Dividing the five proved derivative bounds by z_x and integrating therefore gives $|\mathcal{H}_{a_q}(F(1)) - \mathcal{H}_{a_q}(F(0))| \leq I_q$. Moreover $I_q \leq \int dx = \log(b_n/b_1)$.

For every integer $p \geq 0$ retain the following specified finite signed construction, on the same source operators:

$$\begin{aligned} B_x^\circ &= (1-x)I + xD, & \alpha_x &= \tfrac{1}{2}(2(1-x) + x(b_1 + b_n)), & H_x^\circ &= I - B_x^\circ/\alpha_x, \\ C_x^{[p]} &= \alpha_x^{-1} \sum_{r=0}^p (H_x^\circ)^r (D - I), \\ R_x^{[p]} &= (H_x^\circ)^{p+1} C_x, & E_x^{[p]} &= R_x^{[p]} - \operatorname{Tr}(R_x^{[p]})I/n, \\ j_p &= \int_0^1 \operatorname{Tr}(C_x^{[p]}(U_x - W_x)) dx, \\ e_p &= \int_0^1 \sqrt{\operatorname{Tr}((E_x^{[p]})^2) \operatorname{Tr}((U_x - W_x)^2)} dx. \end{aligned} \quad (\text{ACM15})$$

The finite geometric identity proves $C_x - C_x^{[p]} = R_x^{[p]}$. Every factor is a real function of D , hence self-adjoint in M_x . Trace zero removes the centered scalar, so Hilbert–Schmidt Cauchy–Schwarz proves $j_p - e_p \leq \Delta_{h,k} \leq j_p + e_p$. Writing $\vartheta = (b_n - b_1)/(b_n + b_1) < 1$ and $K_D = \sum_{j=1}^n (|b_j - 1|/\min\{1, b_j\})^2$, the full residual eigenvalues give $\text{Tr}((E_x^{[p]})^2) \leq \vartheta^{2p+2} K_D$. Both U_x, W_x dominate their range projections, whose intersection has dimension at least one, so $\text{Tr}(U_x W_x) \geq 1$. Equation (ACM10) then gives $e_p \leq \vartheta^{p+1} \sqrt{K_D(8q-6)} \rightarrow 0$. This retains the exact signed center and its controlled residual, with no assumption of uniformity in the tensor order.

Put $F_0 = F(0)$ and $H_0 = \mathcal{H}_{a_q}(F_0)$. For every $p \geq 0$ define the actual simultaneous endpoint functions

$$\begin{aligned}\beta_p^- &= \max\{0, F_0 - J_q, \Psi_{a_q}(\max\{0, H_0 - I_q\}), F_0 + j_p - e_p\}, \\ \beta_p^+ &= \min\{F_0 + J_q, \Psi_{a_q}(H_0 + I_q), F_0 + j_p + e_p\}.\end{aligned}\tag{ACM16}$$

The preceding additive, nonlinear and signed inequalities prove $\beta_p^- \leq F(1) \leq \beta_p^+$ by taking their intersection. Both original angle lists, source masses, relation maps and arithmetic Taylor unit remain fixed. At $D = cI$ all diameter and residual terms vanish: each quotient volume is multiplied by the same scalar power q , which cancels with the four signs. The formulas then give $F(1) = F_0$ exactly. Numerical integration of a signed center must retain its own integration error; (ACM15) specifies exact integrals.

At every independent coefficient face $(W, S; A)$ the coefficient inclusions and zero entries act componentwise on these same maps. The original minimum-section difference is a relation χQ ; successive monic division constructs $\chi(\sum s_i)Q(\sum s_i) = \sum_i h(s_i)Q_i(s)$. Its original primitive is $\sum_i (-1)^{i-1} Q_i(D_1, \dots, D_k)(F_h^{\otimes(i-1)} \otimes \phi_* \otimes F_h^{\otimes(k-i)})$. The tensor differential contributes the same sign and therefore recovers the relation. At a proper declared support its residual is the class in $(d\mathcal{C}_{\text{full}}^{k-1} \cap \mathcal{C}_{W,S;A}^k)/d\mathcal{C}_{W,S;A}^{k-1}$, with the inclusion-induced map and inverse given by the same representative, as proved in (ISM38)–(ISM43). Consequently none of the metric estimates deletes this residual, the full diagonal degree-zero source, or an empty mask's outer label.

2 Actual bounds for the retained four-window costs

Continue with (HT1)–(HT18), now with $k \geq 3$. All unadorned source norms and quotient volumes in this section are those of the original $m_{h,k}(u) du$, with $g = 2\xi$ and the complete hypothetical quartet. Write $\mathcal{L} = \mathcal{L}_{h,k}$ and retain $q = \ell_k(k+1)^2$. The threshold-four lower bound is inherited from *Consecutive identities and the first window*, (CJ.9)–(CJ.19); it is already explicitly present in the recovered transcript at lines 4202–4228. It is not a new claim of the present continuation.

2.1 Every term of the inherited lower bound

Put $d_j = V_j/V_{j-1}$ for $j \geq q$. For the two original windows $I_- = (q-1, 2q-1)$ and $I_+ = (q, 2q)$, retain the contraction penalties

$$\begin{aligned}P_- &= -\log(1 - d_{2q-1}) - 2 \sum_{j=q}^{2q-2} \log(1 - d_j), \\ P_+ &= -\log(1 - d_q) - \log(1 - d_{2q}) - 2 \sum_{j=q+1}^{2q-1} \log(1 - d_j).\end{aligned}\tag{HC1}$$

The source identities prove $0 < d_j < 1$ in these ranges. The real phase ϕ_N is defined by the retained source pairing $b_N^* G_N b_{N+1} = i\omega_N \phi_N$. The first window uses its original boundary identity;

no inverse at degree $q - 2$ occurs. Define weights

$$w_{q-1} = w_{2q-1} = 1, \quad w_N = 2 \quad (q \leq N \leq 2q - 2), \quad \sum_{N=q-1}^{2q-1} w_N = 2q.$$

Adding the two exact source products gives

$$\mathcal{B}_k := \log \frac{V_{q-1} V_q}{V_{2q-1} V_{2q}} = \sum_{N=q-1}^{2q-1} w_N \log(\varepsilon_N^2 + \phi_N^2) - \log \frac{\omega_{2q-1} \omega_{2q}}{\omega_{q-1} \omega_q} + P_- + P_+. \quad (\text{HC2})$$

Here every $\varepsilon_N \geq \mathcal{L}$, including the first degree.

For clarity, retain all constants in the inherited comparable-window bound. With $\alpha = \pi/2$, $B = 42 + 8m$, and $\vartheta_h = \int_{-1}^1 w_h$, put

$$\begin{aligned} t_h &= \min(1, \vartheta_h), \quad c_0 = \min(1, 2c_h/3), \quad \ell_h = \frac{1}{2} e^{-\alpha} \sqrt{c_0 t_h} 2^{-B/2}, \\ X &= \max(1, M_h(b), M_h(-b)), \quad U_h = 4\sqrt{X}/b, \\ C_h^{\text{win}} &= 27 \max(1, U_h)^3 \max(1, \ell_h^{-1})^2, \quad D_h = \frac{\delta}{2C_h^{\text{win}}}. \end{aligned} \quad (\text{HC3})$$

Here $0 < b < \pi/2$, and $c_h > 0$ is the original proved convolution constant. The source proves $(\omega_{n+r}/\omega_n)^{1/(2r)} \leq C_h^{\text{win}} n$ for $n \geq k \geq 3$, $n/2 \leq r \leq 2n$. Thus it applies to both literal windows, and $0 < D_h < 1/108$.

The exact residual in (HC2) has the following five nonnegative pieces:

$$\begin{aligned} \mathcal{R}_k &= P_- + P_+ + \sum_N w_N \log(1 + \phi_N^2/\mathcal{L}^2) \\ &\quad + \sum_N w_N \log \frac{\varepsilon_N^2 + \phi_N^2}{\mathcal{L}^2 + \phi_N^2} \\ &\quad + 4q \log(C_h^{\text{win}} q) - \log \frac{\omega_{2q-1} \omega_{2q}}{\omega_{q-1} \omega_q} + 4q \log \frac{2\mathcal{L}}{\delta k q}. \end{aligned} \quad (\text{HC4})$$

Both sums use exactly $q - 1 \leq N \leq 2q - 1$. Nonnegativity follows, respectively, from $0 < d_j < 1$, the squared phases, (HT9), the two comparable-window estimates, and $\mathcal{L}/q \geq \delta k/2$. Expanding each logarithm in (HC4), with $D_h = \delta/(2C_h^{\text{win}})$, proves the exact identity

$$\boxed{\mathcal{B}_k = 4q \log(D_h k) + \mathcal{R}_k, \quad \mathcal{R}_k \geq 0.} \quad (\text{HC5})$$

This retains, rather than discards, all the contraction, phase, norm, and trace savings in the original lower bound.

2.2 A proved upper bound for these same four original volumes

The original convolution estimate and Laplace moments are

$$m_{h,k}(u) \geq c_h \vartheta_h^{k-3} e^{-\alpha(|u|+k-3)} (1 + |u| + k - 3)^{-B}, \quad M_h(\pm b) = \int_{\mathbb{R}} e^{\pm bu} w_h(u) du.$$

Write $\mu_h = \int_{\mathbb{R}} w_h(u) du$ for the original source mass. Put

$$\begin{aligned} R_0 &= \sqrt{\delta^2 + \gamma^2}, \quad D = \max(1, 2/b, 2R_0), \quad T = Dq, \quad r = kR_0/T \leq 1/2, \\ a_k(T) &= c_h \vartheta_h^{k-3} e^{-\alpha(T+k-3)} (1 + T + k - 3)^{-B}, \\ M_k &= M_h(b)^k + M_h(-b)^k, \quad U_k(T) = \mu_h^k + \frac{(2q-2)!}{(bT)^{2q-2}} M_k, \\ A_N &= \sqrt{\frac{2N+1}{2}} \frac{4^{N+1} - 1}{3}, \quad B_q(r) = \left(\frac{1+r}{1-r} \right)^q, \quad C_N = \frac{U_k(T) A_N^2 B_q(r)^2}{T a_k(T)}. \end{aligned} \quad (\text{HC6})$$

The capital D in the interval scale is distinct from the inherited lower constant D_h and from the source differential. All masses, coordinates, and root multiplicities remain as in (HT1)–(HT7). One has the actual finite Hermitian comparisons

$$G_{q-1} \preceq C_{2q-1} G_{2q-1}, \quad G_q \preceq G_{q-1} \preceq C_{2q} G_{2q}.$$

Consequently, with the exact ratio

$$\rho_q = \frac{4q+1}{4q-1} \left(\frac{4^{2q+1}-1}{4^{2q}-1} \right)^2,$$

$$\boxed{0 \leq \mathcal{B}_k \leq q \log C_{2q-1} + q \log C_{2q} = 2q \log C_{2q-1} + q \log \rho_q.} \quad (\text{HC7})$$

Proof of the degree extension and its maps. Use the exact substitution $f(x) = P(c + iTx)$ with inverse $P(S) = f((S - c)/(iT))$. Its source measure is $Tm_{h,k}(Tx) dx$, with mass μ_h^k , and its monic relation is $\chi_T(x) = (iT)^{-q} \chi(c + iTx)$. The roots $z_1, \dots, z_q$, listed with their repetitions, have modulus at most r . The raw jet of order d acquires exactly $(iT)^d$; the quotient determinant acquires $T^{-q(q-1)}$. This common factor cancels in each displayed quotient-volume ratio.

The Legendre expansion on $[-1, 1]$ gives $\|f\|_{\text{coeff},1} \leq A_N \|f\|_{L^2([-1,1])}$ for $\deg f \leq N$. Indeed Rodrigues' polynomial has coefficient norm at most $\binom{2j}{j} \leq 4^j$ and squared integral $2/(2j+1)$. Pairing the expansion with this polynomial bounds its coefficient by $\sqrt{(2j+1)/2} \|f\|_2$; summing the geometric series gives exactly A_N in (HC6).

Write $\prod_a (1 - z_a t)^{-1} = \sum_{j \geq 0} h_j t^j$. Geometric-series multiplication gives $|h_j| \leq \binom{q+j-1}{j} r^j$ and $\|\chi_T\|_{\text{coeff},1} \leq (1+r)^q$. For every integer $s \geq 0$, monic division of x^{q+s} has quotient $\sum_{j=0}^s h_j x^{s-j}$. The inverse-series identity cancels all other terms of degree at least q . Removing the leading coefficient one therefore gives

$$\|\text{rem}_{\chi_T} x^{q+s}\|_{\text{coeff},1} \leq (1+r)^q \sum_{j=0}^s \binom{q+j-1}{j} r^j - 1 \leq B_q(r).$$

The bound is valid also for $s = q$, the extra degree required here. At degrees below q the remainder is the original monomial, whose coefficient norm is one. Linearity gives the same remainder-map bound for every degree at most $2q$.

For a remainder v of degree below q ,

$$\int |v(u/T)|^2 m_{h,k}(u) du \leq U_k(T) \|v\|_{\text{coeff},1}^2.$$

This follows by bounding each monomial by $\max(1, |u/T|^{q-1})$ and using $\int u^{2q-2} m_{h,k} \leq (2q-2)! b^{-(2q-2)} M_k$. Restriction of the original norm to $[-T, T]$ gives the lower bound $T a_k(T) \|f\|_{L^2([-1,1])}^2$. Combining these three estimates proves

$$\|\text{rem}_{\chi} P\|_{H_{q-1}}^2 \leq C_N \|P\|_{H_N}^2 \quad (N = 2q-1, 2q).$$

The map is the literal remainder representative of the original jet. Apply it to the canonical least lift of $x \in E$ at degree N . Its norm at degree $q-1$ is at least $x^* G_{q-1} x$; this proves the Hermitian comparisons preceding (HC7). Inclusion of polynomial degrees gives $G_q \preceq G_{q-1}$ and monotonicity of all volumes. Taking determinants proves (HC7). Applying the same remainder inequality to any nonzero polynomial below degree q proves $C_N \geq 1$ without an additional assumption. $\square$

For an entirely scalar upper bound retain the original constants

$$K = \max(1, \vartheta_h^{-1}), \quad C_{\text{env}} = \max(1, 2/(3c_h D)), \quad a = \alpha D + \log 256,$$

and the original expression (AU.31)

$$\mathcal{U}_k = aq^2 + \left(\log(XK) + \alpha + \frac{16R_0}{3D} \right) kq + Bq \log(1 + (D+1)q) + q \log C_{\text{env}}.$$

Then the proved four-window upper bound is

$$\boxed{\mathcal{B}_k \leq \widehat{\mathcal{U}}_k := 2\mathcal{U}_k + q \log 20.} \quad (\text{HC8})$$

To verify the constants, $bT \geq 2q$ implies the factorial ratio in (HC6) is at most one; also $\mu_h \leq X$ and $M_k \leq 2X^k$, so $U_k(T) \leq 3X^k$. Directly from (HC6),

$$A_{2q-1}^2 \leq (2q/9)256^q, \quad A_{2q}^2 \leq (40q/9)256^q.$$

The logarithmic derivative $2/(1-r^2) \leq 8/3$ on $[0, 1/2]$ gives $\log B_q(r)^2 \leq 16kR_0/(3D)$. Substitution into C_N yields $q \log C_{2q-1} \leq \mathcal{U}_k$ and $q \log C_{2q} \leq \mathcal{U}_k + q \log 20$. Adding proves (HC8). This argument does not use the false inequality $\rho_2 < 20$; in the actual quartet domain $q \geq 16$ anyway.

2.3 The exact compatible interval for the signed arithmetic correction

Let V_N^Γ denote the inherited Gamma-reference quotient volume for the *same* original polynomial χ , and define

$$T_N = V_N/V_N^\Gamma, \quad \mathcal{B}_k^\Gamma = \log \frac{V_{q-1}^\Gamma V_q^\Gamma}{V_{2q-1}^\Gamma V_{2q}^\Gamma}, \quad \Delta_k^\Gamma = \log \frac{T_{q-1} T_q}{T_{2q-1} T_{2q}}.$$

Cancellation of the four literal volume factors gives $\mathcal{B}_k = \mathcal{B}_k^\Gamma + \Delta_k^\Gamma$. No auxiliary polynomial is introduced and no sign is imposed on Δ_k^Γ . The exact coordinate identification with (ACM1)–(ACM5) is $y = u$, with $S = k/2 + iu$ and unchanged ascending coefficients. Its arithmetic form is the original $m_{h,k} = w_h^{*k}$, while the reference form is precisely (ACM2), including its full convolution mass. The relation matrices are the same multiplication by the full χ of (HT2). Thus apply (ACM11a)–(ACM16) to this pair and write $\beta_{p,\Gamma}^\pm$ for the resulting simultaneous endpoints, whose initial value is $F_0 = \mathcal{B}_k^\Gamma$. In particular $\beta_{p,\Gamma}^- \leq \mathcal{B}_k \leq \beta_{p,\Gamma}^+$. This uses every spectral, trace-norm, overlap, variance, angle and constructed signed-residual control on the original Gamma-to-arithmetic path. Put $T_k = 4q \log(D_h k)$ for this scalar expression only. Then (HC5), (HC8) and these endpoints give the current intervals

$$\boxed{\max\{0, -T_k, \beta_{p,\Gamma}^- - T_k\} \leq \mathcal{R}_k \leq \min\{\widehat{\mathcal{U}}_k, \beta_{p,\Gamma}^+\} - T_k.} \quad (\text{HC9})$$

$$\begin{aligned} \max\{0, T_k, \beta_{p,\Gamma}^-\} - \mathcal{B}_k^\Gamma &\leq \Delta_k^\Gamma \\ &\leq \min\{\widehat{\mathcal{U}}_k, \beta_{p,\Gamma}^+\} - \mathcal{B}_k^\Gamma. \end{aligned} \quad (\text{HC10})$$

Every entry follows by subtracting the same displayed initial value from a proved bound on $\mathcal{B}_k$. Removing the source-dependent $\beta_{p,\Gamma}^\pm$ entries gives the earlier coarse intervals. Those coarse intervals have strictly positive width for every $k \geq 3$.

Indeed all nonquadratic terms of $\mathcal{U}_k$ are nonnegative, $a > 1$, $q \geq (k+1)^2 > 2k$, and $D_h < 1$. Thus

$$\widehat{\mathcal{U}}_k \geq 2aq^2 > 4qk \geq \max\{0, 4q \log(D_h k)\}.$$

Moreover, keeping the fixed packet constants, each lower-order term divided by q^2 tends to zero, so

$$\lim_{k \rightarrow \infty} \frac{\widehat{\mathcal{U}}_k - \max\{0, 4q \log(D_h k)\}}{q^2} = 2(\alpha D + \log 256) > 0. \quad (\text{HC11})$$

This proves the stated width of the coarse bound functions obtained by removing the source-dependent entries in (HC9)–(HC10). The refined intersection retains those computed entries; (HC11) makes no assertion that its width equals the coarse width. It does not construct a packet realizing an arbitrarily chosen point of either interval, and it does not assert the existence of the hypothetical zero. In particular the interval may lie entirely at negative values of Δ_k^Γ ; determinant-ratio positivity does not imply positivity of its signed four-window logarithm.

2.4 The holonomy energy is tied to the same endpoint contractions

Fix an actual holonomy θ and period L . In this paragraph all metrics, lifts and volumes carry that same phase. Put

$$e_N = \log(V_N/V_{N+1}), \quad \beta_N = \lambda_{\max}(G_{N+1}^{-1/2} G_N G_{N+1}^{-1/2}),$$

$$J_N^D = \|D_{L,\theta} R_N G_N^{-1/2}\|_{\text{HS}}^2, \quad \mathcal{E}_N = \|\mathcal{B}_N G_N^{-1/2}\|_{\text{HS}}^2.$$

These use original source moments through degree $2N + 2$. They obey

$$\boxed{\frac{\mathcal{L}^2}{2} \leq \mathcal{E}_N \leq (1 + \sqrt{\beta_N})^2 J_N^D, \quad e_N \geq 2 \log \max \left\{ 1, \frac{\mathcal{L}}{\sqrt{2J_N^D}} - 1 \right\}}. \quad (\text{HC12})$$

The denominator is positive: integration by parts on the stated twisted domain gives $\|D_{L,\theta} f\|^2 = c^2 \|f\|^2 + \|\partial_r f\|^2$, whence $J_N^D \geq c^2 q > 0$.

To prove the upper inequality, $D_{L,\theta} R_N v$ is an original degree- $(N + 1)$ representative of Av . The same quotient minimum therefore gives

$$A^* G_{N+1} A \preceq R_N^* D_{L,\theta}^* D_{L,\theta} R_N.$$

Use $G_N \preceq \beta_N G_{N+1}$, sandwich by $G_N^{-1/2}$, and take traces. This gives $\|R_N A G_N^{-1/2}\|_{\text{HS}}^2 \leq \beta_N J_N^D$. The triangle inequality in $\mathcal{B}_N = D_{L,\theta} R_N - R_N A$ proves the upper inequality in (HC12); (HT14) proves the lower. All eigenvalues in the definition of β_N are at least one by degree monotonicity. In fact the original rank-one kernel update proves the exact identity

$$e_N = \log \beta_N, \quad \beta_N = 1 + \frac{b_{N+1}^* G_N b_{N+1}}{\omega_{N+1}}. \quad (\text{HC12a})$$

Here b_{N+1} and ω_{N+1} are the remainder column and monic squared norm for this same original phase measure. Indeed $K_{N+1} = K_N + b_{N+1} b_{N+1}^* / \omega_{N+1}$, so

$$K_N^{-1/2} K_{N+1} K_N^{-1/2} = I + vv^*, \quad v = K_N^{-1/2} b_{N+1} / \sqrt{\omega_{N+1}}.$$

The relative matrix defining β_N is $K_{N+1}^{-1/2} K_N^{-1} K_{N+1}^{-1/2} = X^* X$ for the invertible $X = K_N^{-1/2} K_{N+1}^{1/2}$. The displayed matrix $I + vv^*$ is $X X^*$, and $X(X^* X)X^{-1} = X X^*$ proves they have identical eigenvalues. These are 1 with multiplicity at least $q - 1$ and $1 + \|v\|^2$. Taking the determinant gives $V_N/V_{N+1} = \det K_{N+1} / \det K_N = 1 + \|v\|^2 = \beta_N$, which proves (HC12a), including a zero update if one occurs. Rearranging the two energy bounds now proves the second inequality of (HC12) with this exact endpoint identity.

The same weights as in (HC2) reconstruct the actual phase volume:

$$\boxed{\mathcal{B}_{k,\theta} = \sum_{N=q-1}^{2q-1} w_N e_N \geq 2 \sum_{N=q-1}^{2q-1} w_N \log \max \left\{ 1, \frac{\mathcal{L}}{\sqrt{2J_N^D}} - 1 \right\}}. \quad (\text{HC13})$$

Thus a small original derivative-source energy at a degree has the explicit endpoint contraction cost (HC12); that cost enters the same four-window budget, with its actual multiplicity.

Finally the supplied holonomy construction allows any fixed $0 < \eta < 1$ by choosing the original period L from its computed weighted source comparison at degree $2q + 2$. We now apply that comparison on the actual source before quotient minimization. Let $I_{2q}^{2q+2} : \mathcal{P}_{2q} \hookrightarrow \mathcal{P}_{2q+2}$ be the literal coefficient inclusion and retain $H = \mathcal{P}_{2q}$. The exact forms on H are

$$M_0^\theta = (I_{2q}^{2q+2})^* M_{2q+2} I_{2q}^{2q+2} = \int v(u)^* v(u) m_{h,k}(u) du,$$

$$M_1^\theta = (I_{2q}^{2q+2})^* M_{2q+2,\theta} I_{2q}^{2q+2} = \frac{2\pi}{L} \sum_{r \in \mathbb{Z}} m_{h,k}(u_r) v(u_r)^* v(u_r), \quad (\text{HC14a})$$

$$v(u) = (1, c + iu, \dots, (c + iu)^{2q}), \quad c = k/2, \quad u_r = (2\pi r + \theta)/L.$$

This is (HT6) with its original mass and lattice. In terms of the actual source maps the two forms are the squared norms of $\mathcal{U}_k\mathcal{V}|_H$ and $\mathcal{Z}_{L,\theta}\mathcal{U}_k\mathcal{V}|_H$, respectively. Thus no relative source variable, full Taylor unit or coefficient is removed. Restricting the proved (PSC.33) source inequality by I_{2q}^{2q+2} proves $(1 - \eta)M_0^\theta \preceq M_1^\theta \preceq (1 + \eta)M_0^\theta$. For each polynomial degree $q - 1 \leq N \leq 2q$, both quotient minima are over the identical fibre $J_N P = v$; the sandwich also gives the earlier quotient comparisons by taking infima on those fibres.

Set $M_x^\theta = (1 - x)M_0^\theta + xM_1^\theta$ for $x \in [0, 1]$. Use the literal $I_N : \mathcal{P}_N \rightarrow H$ and the unpadded relation $B_N^{\text{AW}} : \mathcal{P}_{N-q} \rightarrow \mathcal{P}_N$, $P \mapsto \chi P$, in (ACM4), with $M_x = M_x^\theta$. Thus the full relation map is $B_N^\theta = I_N B_N^{\text{AW}} : \mathcal{P}_{N-q} \rightarrow H$ and the resulting projections are exactly

$$\begin{aligned} P_N^\theta &= I_N (I_N^* M_x^\theta I_N)^{-1} I_N^* M_x^\theta, \\ Q_N^\theta &= B_N^\theta ((B_N^\theta)^* M_x^\theta B_N^\theta)^{-1} (B_N^\theta)^* M_x^\theta. \end{aligned}$$

The negative relation domain is the zero space. Put

$$\begin{aligned} U_x^\theta &= Q_{2q-1}^\theta + Q_{2q}^\theta - Q_q^\theta, \\ W_x^\theta &= P_{2q-1}^\theta - P_{q-1}^\theta + P_{2q}^\theta - P_q^\theta, \\ C_x^\theta &= (M_x^\theta)^{-1} (M_1^\theta - M_0^\theta), \quad D_\theta = (M_0^\theta)^{-1} M_1^\theta. \end{aligned} \tag{HC14b}$$

Differentiating each canonical quotient Gram exactly as in (ISM18) gives $F'_\theta(x) = \text{Tr}(C_x^\theta (U_x^\theta - W_x^\theta))$, where $F_\theta(0) = \mathcal{B}_k$ and $F_\theta(1) = \mathcal{B}_{k,\theta}$. In particular the zero relation domain still gives $Q_{q-1}^\theta = 0$. The two complete spectra are $0^{[q]}, 1^{[2]}, 2^{[q-1]}$. The original monomial and relation flags, and all minimum-section angle entries including zeroes, are exactly the ones used in the proof of (ACM11a).

Define $J_\theta, I_\theta, j_{p,\theta}, e_{p,\theta}$ by the literal substitution

$$(M_x, C_x, U_x, W_x, D, F(x)) \mapsto (M_x^\theta, C_x^\theta, U_x^\theta, W_x^\theta, D_\theta, F_\theta(x))$$

throughout (ACM11a) and (ACM15). The index $p \geq 0$ is the finite-series degree; the original period remains L . These are separately calculated source-phase quantities. Explicitly, J_θ is the integral of the pointwise minimum of the full ordered-spectrum expression, the exact midpoint trace norm, the actual-overlap majorant, the centered Hilbert–Schmidt expression, and the full-angle expression; I_θ divides these bounds by $(2q - 1)\sqrt{1 - e^{-F_\theta(x)/(2q-1)}}$ as in (ACM11a). The original continuous measure and the positive lattice measure both have positive Grams, as proved in (HT6). Therefore the $\chi^{\#_k}$ argument in (ACM11a) proves $F_\theta(x) > 0$ on this entire path, including its atomic endpoint. All substitutions in the derivative and nonlinear calculations are consequently valid. The signed construction retains the exact finite residual and gives

$$\begin{aligned} j_{p,\theta} - e_{p,\theta} &\leq \mathcal{B}_{k,\theta} - \mathcal{B}_k \leq j_{p,\theta} + e_{p,\theta}, \\ 0 \leq e_{p,\theta} &\leq \left(\frac{b_n^\theta - b_1^\theta}{b_n^\theta + b_1^\theta} \right)^{p+1} \sqrt{K_{D_\theta}(8q - 6)} \rightarrow 0, \quad n = 2q + 1. \end{aligned} \tag{HC14c}$$

Here every b_j^θ is an eigenvalue of D_θ in M_0^θ , and $K_{D_\theta} = \sum_{j=1}^n (|b_j^\theta - 1| / \min\{1, b_j^\theta\})^2$. The finite geometric identity and the original Hilbert–Schmidt trace proof in (ACM15) prove every term of (HC14c) after the explicit substitution (HC14b). The source sandwich places each b_j^θ in $[1 - \eta, 1 + \eta]$. The simultaneous original phase error is therefore

$$\begin{aligned} |\mathcal{B}_{k,\theta} - \mathcal{B}_k| &\leq J_\theta \leq \mathcal{L}_q(D_\theta) \leq (2q - 1) \log \frac{1 + \eta}{1 - \eta}, \\ |\mathcal{H}_{2q-1}(\mathcal{B}_{k,\theta}) - \mathcal{H}_{2q-1}(\mathcal{B}_k)| &\leq I_\theta \leq \log \frac{1 + \eta}{1 - \eta}. \end{aligned} \tag{HC14}$$

Thus the earlier $2q$ determinant-only allowance follows as a larger bound; every stronger pointwise control and signed center has been retained before any scalar allowance is taken.

For any integers $p, r \geq 0$, put

$$\begin{aligned} U_{k,p}^{\text{ar}} &= \min\{\widehat{\mathcal{U}}_k, \beta_{p,\Gamma}^+\}, \\ U_{k,\theta;p,r} &= \min\{U_{k,p}^{\text{ar}} + J_\theta, \Psi_{2q-1}(\mathcal{H}_{2q-1}(U_{k,p}^{\text{ar}}) + I_\theta), U_{k,p}^{\text{ar}} + j_{r,\theta} + e_{r,\theta}\}. \end{aligned} \quad (\text{HC14d})$$

The actual $\mathcal{B}_k$ is positive and at most $U_{k,p}^{\text{ar}}$ by (HC10). The additive and signed expressions in (HC14d) increase with their initial value. Both $\mathcal{H}_{2q-1}$ and Ψ_{2q-1} increase on the indicated nonnegative domains, so the nonlinear expression does too. Apply the three upper inequalities to F_θ , then substitute this proved initial upper bound. This proves the propagated budget at the original energy use site:

$$2 \sum_{N=q-1}^{2q-1} w_N \log \max \left\{ 1, \frac{\mathcal{L}}{\sqrt{2J_N^D}} - 1 \right\} \leq \mathcal{B}_{k,\theta} \leq U_{k,\theta;p,r}. \quad (\text{HC13a})$$

The w_N , J_N^D and every original derivative-source and boundary term in (HC12)–(HC13) remain unchanged. Coefficient zero insertion at a specified $(W, S; A)$ commutes with the displayed inclusions and linear source maps. The minimum-section comparison retains the original monic-division primitive and the proper-source residual proved in (ISM38)–(ISM43); no residual class or empty mask's outer label is set to zero by this metric calculation. For the averaged original phase metrics the already proved stronger comparison $(1 - \eta^2)G_N \preceq \overline{G}_N \preceq G_N$ gives instead $|\overline{\mathcal{B}}_k - \mathcal{B}_k| \leq 2q \log(1/(1 - \eta^2))$. These changes are of order q for a fixed η , whereas the compatible width (HC11) has positive quadratic order. The exact Laplacian residual (HT17) remains present at every phase. No displayed upper estimate for its norm, or for the original derivative moments in (HC12), contradicts these lower bounds.

2.5 The sharper finite comparison retains its exact error

The original lower-envelope measure is

$$d\nu_k(u) = c_h \vartheta_h^{k-3} e^{-\alpha(k-3)} e^{-\alpha|u|} (k-2+|u|)^{-B} du.$$

Let V_N^ν be its quotient volume for the identical full remainder map. The actual finite upper expression for the four-window volume is

$$\mathcal{U}_{\nu,k} = \log(V_{q-1}V_q) - \log(V_{2q-1}^\nu V_{2q}^\nu),$$

with the exact nonnegative error

$$\boxed{\mathcal{U}_{\nu,k} - \mathcal{B}_k = \log \frac{V_{2q-1}}{V_{2q-1}^\nu} + \log \frac{V_{2q}}{V_{2q}^\nu} \geq 0.} \quad (\text{HC15})$$

Indeed $m_{h,k}(u) du \geq d\nu_k$ bounds every source polynomial quadratic form. Minimizing on each identical affine remainder fibre gives $G_N \succeq G_N^\nu \succ 0$. Conjugation by $(G_N^\nu)^{-1/2}$ shows that each volume ratio in (HC15) is at least one. Cancellation proves the equality. The low-degree factors require only original moments through $2q$, and the comparison moments in the last factors go through $4q$. These are well-defined finite expressions, with the actual full jet map and source mass. The supplied proof records do not evaluate their errors at the hypothetical arithmetic quartet or bound them at a rate opposing (HC5). Retaining this sharper finite expression therefore supplies no extra contradictory inequality.

3 Recursive propagation of the original signed metric bounds

This proof combines the earlier full-relative-spectrum estimate AW16, the exact trace norm and actual-overlap estimates of SP, the supplied centered Hilbert–Schmidt estimate, and the all-degree

angle estimate EP17–EP25 on their common original source. Each of those complete proofs is retained in this edition. The construction below carries their pointwise minimum through both the original volume and the exact nonlinear endpoint coordinate, without changing either endpoint form.

3.1 The original source and all four quotient volumes

Fix the original packet h , tensor order $k \geq 1$, monic cyclic polynomial $\chi = \chi_{h,k}$ of degree $q \geq 1$, and

$$H = \mathcal{P}_{2q}, \quad S = k/2 + iy, \quad v(y) = (1, S, \dots, S^{2q}). \quad (\text{RMT1})$$

The coordinates are the original ascending monomial coordinates. The original reference and arithmetic measures are

$$d\mu_0 = r_{1/4}^{*k}(y) dy, \quad d\mu_1 = w_h^{*k}(y) dy, \quad w_h(y) = \frac{|(2\xi/h)(1/2 + iy)|^2}{2\pi}. \quad (\text{RMT2})$$

All their original mass and Taylor-unit factors remain. In particular the exact reference convolution identity is

$$r_{1/4}^{*k} = \frac{c_{1/4}^k}{c_{k/4}} r_{k/4}, \quad c_a = 2^{1-2a} \Gamma(2a), \quad c_{1/4} = \sqrt{2\pi}. \quad (\text{RMT3})$$

Thus these are the same measures in AW and SP. Every argument below also applies to the actual finite-circle comparison when one of the measures is its specified positive atomic measure on this same line and its retained degree- $2q$ Gram is positive definite. That positivity is already proved by the original sampled-source bounds. This statement does not assign moment structure to an arbitrary positive coefficient matrix.

For $0 \leq x \leq 1$ let

$$M_x = \int v(y)^* v(y) d((1-x)\mu_0 + x\mu_1)(y), \quad \dot{M} = M_1 - M_0, \quad C_x = M_x^{-1} \dot{M}. \quad (\text{RMT4})$$

The two positive endpoint forms make M_x positive definite throughout the closed interval. For every original cutoff N , let $I_N : \mathcal{P}_N \rightarrow H$ be coefficient inclusion and $B_N : \mathcal{P}_{N-q} \rightarrow H$ multiplication by χ , followed by inclusion. Negative relation degree means the zero space. Retain the orthogonal projections

$$P_N = I_N(I_N^* M_x I_N)^{-1} I_N^* M_x, \quad Q_N = B_N(B_N^* M_x B_N)^{-1} B_N^* M_x, \quad Q_{q-1} = 0. \quad (\text{RMT5})$$

AW's relation matrix has target $\mathcal{P}_N$, so its exact relation to this one is $B_N = I_N B_N^{\text{AW}}$. Equation (RMT5), after that substitution, is precisely AW4 and SP2. The quotient map remains $J_N : \mathcal{P}_N \rightarrow E = \mathbb{C}[S]/(\chi)$, and its canonical positive quotient Gram is $G_N(x)$. Put $V_N(x) = \det G_N(x)$ and retain the four signs:

$$\mathcal{B}(x) = \log \frac{V_{q-1}(x) V_q(x)}{V_{2q-1}(x) V_{2q}(x)}, \quad U_x = Q_{2q-1} + Q_{2q} - Q_q, \quad W_x = P_{2q-1} - P_{q-1} + P_{2q} - P_q. \quad (\text{RMT6})$$

The relation-first determinant-line formula and differentiation of each of these four determinants give, without commuting their operators,

$$\mathcal{B}'(x) = \text{Tr}((U_x - W_x)C_x). \quad (\text{RMT7})$$

This is the same actual integrand as AW6 and SP7.

3.2 The simultaneous bounds on that integrand

Let $b_1 \leq \dots \leq b_{2q+1}$ be every eigenvalue, with multiplicity, of $D = M_0^{-1}M_1$ in the original M_0 metric. They are positive. Retain

$$c_j(x) = \frac{b_j - 1}{1 - x + xb_j}, \quad d_x = c_{2q+1}(x) - c_1(x), \quad \kappa = \frac{b_{2q+1}}{b_1}. \quad (\text{RMT8})$$

These are the ordered eigenvalues of C_x . Indeed $C_x = ((1-x)I + xD)^{-1}(D - I)$, and differentiation with respect to b_j gives the positive quantity $(1 - x + xb_j)^{-2}$. In particular $\int_0^1 d_x dx = \log \kappa$.

Set $s_x = \dim(\text{ran } U_x \cap \text{ran } W_x)$. The two ranks are $q + 1$ in dimension $2q + 1$, so $s_x \geq 1$. All traces and trace norms below are in the same original M_x metric. Define the two calculated functions

$$\begin{aligned} E_q(x) &= c_{q+2}(x) - c_q(x) + 2 \sum_{j=1}^{q-1} (c_{2q+2-j}(x) - c_j(x)), \\ O_q(x) &= \frac{d_x}{2} \min \left\{ 4q - 2s_x, \sqrt{(2q+1)(8q-4-2\text{Tr}(U_x W_x))} \right\}. \end{aligned} \quad (\text{RMT9})$$

The sum is empty when $q = 1$. The complete nested-flag calculation gives both U_x, W_x the spectrum $0^{[q]}, 1^{[2]}, 2^{[q-1]}$. Decomposing each as the sum of its spectral projections of ranks $q + 1$ and $q - 1$, and applying the ordered projection-trace inequalities, gives $|\mathcal{B}'| \leq E_q$, exactly as in AW15–AW16. For clarity the subtraction of the two bounds cancels the common c_{q+1} only after both sums have been written; it leaves every term displayed in E_q .

Both positive operators dominate the original orthogonal projection onto their common range. Subtracting that same projection from both leaves positive operators with traces $2q - s_x$, whence $\|U_x - W_x\|_1 \leq 4q - 2s_x$. Also their computed spectra give

$$\text{Tr}((U_x - W_x)^2) = 8q - 4 - 2\text{Tr}(U_x W_x). \quad (\text{RMT10})$$

Cauchy–Schwarz on all $2q + 1$ eigenvalues of $U_x - W_x$ gives the second trace-norm bound in (RMT9). Since $\text{Tr}(U_x - W_x) = 0$, subtract $(c_1 + c_{2q+1})I/2$ from C_x in (RMT7). In an M_x -orthonormal eigenbasis of $U_x - W_x$, the absolute value of each diagonal entry of this centered C_x is at most $d_x/2$. Multiplying by the corresponding absolute eigenvalues and summing gives the retained exact trace-norm bound

$$|\mathcal{B}'| \leq T_q(x) := \frac{d_x}{2} \|U_x - W_x\|_1 \leq O_q(x). \quad (\text{RMT10a})$$

The second inequality follows from the two trace-norm estimates just proved. Thus neither of those majorants replaces the original trace norm.

The user's additional supplied projector calculation gives another bound on this same integrand. Define

$$\Sigma_x = \text{Tr}(C_x^2) - \frac{(\text{Tr } C_x)^2}{2q+1}, \quad S_q(x) = \sqrt{\Sigma_x \text{Tr}((U_x - W_x)^2)}. \quad (\text{RMT10b})$$

The nonnegative number Σ_x is the squared Hilbert–Schmidt norm of $C_x - \text{Tr}(C_x)I/(2q+1)$ in the original M_x Hilbert space. The trace of $U_x - W_x$ is zero, so Hilbert–Schmidt Cauchy–Schwarz gives $|\mathcal{B}'| \leq S_q$. Equivalently one may apply the isometry $M_x^{1/2} : (H, M_x) \rightarrow (H, I)$ simultaneously to both operators; conjugation preserves their traces and products and proves the same inequality. This is a calculation with the original metric, not a change of either endpoint source form.

The original crossed quotient pairs are $(q-1, 2q)$ and $(q, 2q-1)$. Retain their full canonical minimum-section angle lists $(\theta_1, \dots, \theta_q)$ and $(\varphi_1, \dots, \varphi_q)$. In the second list, ordered with a smallest angle first, the original ambient dimension forces $\varphi_1 = 0$. The count is proved in EP17–EP25. For the following scalar sums put $a = 2q - 1$ and define the explicitly indexed list

$$\lambda_i = -\log \cos^2 \theta_i \quad (1 \leq i \leq q), \quad \lambda_{q+j-1} = -\log \cos^2 \varphi_j \quad (2 \leq j \leq q).$$

Then $\mathcal{B} = -\log \cos^2 \varphi_1 + \sum_{\nu=1}^a \lambda_\nu = 0 + \sum_{\nu=1}^a \lambda_\nu$. Every additional zero remains in the λ -list, while the designated geometric entry φ_1 is still displayed in the full second list. The exact projection-difference spectra give

$$|\mathcal{B}'(x)| \leq d_x \left(\sin \varphi_1 + \sum_{\nu=1}^a \sqrt{1 - e^{-\lambda_\nu(x)}} \right) \leq a \sqrt{1 - e^{-\mathcal{B}(x)/a}} d_x. \quad (\text{RMT11})$$

The second inequality is concavity of $z \mapsto \sqrt{1 - e^{-z}}$ on $[0, \infty)$, applied to the full a -term list, including its zeros, together with the displayed $\sin \varphi_1 = 0$. Both q -entry lists remain the geometric data; the known zero contribution is kept explicitly in these scalar identities.

The needed strict positivity of $\mathcal{B}$ follows on the actual measures rather than being an additional assumption. If $\mathcal{B} = 0$, both nested quotient pairs have equal minimum spaces. To see every implication, their canonical minimum sections satisfy $G_i - G_j = (R_i - R_j)^* M_x (R_i - R_j) \succeq 0$. Thus each determinant ratio in the crossed pairs is at least one. A zero sum of their nonnegative logarithms makes both ratios one. All eigenvalues of $G_j^{-1/2} G_i G_j^{-1/2}$ are at least one, so determinant one makes every eigenvalue one and $G_i = G_j$. The displayed squared norm then gives $R_i = R_j$ as maps to the same original source H . In the first pair R_{q-1} is the literal degree-less-than- q remainder section. Its equality with R_{2q} puts the constant polynomial 1 in the latter minimum space, which is orthogonal to $\ker J_{2q} = \chi \mathcal{P}_q$. For $\chi(S) = \sum a_j S^j$, put $\chi^{\#k}(S) = \sum \bar{a}_j (k - S)^j$. This belongs to $\mathcal{P}_q$ and equals $\overline{\chi(S)}$ on $S = k/2 + iy$. The asserted orthogonality would therefore give

$$0 = \int \chi(S) \chi^{\#k}(S) d\mu_x = \int |\chi(S)|^2 d\mu_x, \quad (\text{RMT12})$$

contradicting the positive Gram of the nonzero polynomial χ . This proof also covers the specified positive atomic comparison. Thus $\mathcal{B}(x) > 0$ on the entire closed interval.

3.3 Carrying the refined bound through both endpoint coordinates

Define the positive function

$$z_x = a \sqrt{1 - e^{-\mathcal{B}(x)/a}} > 0.$$

With every preceding control retained, set

$$\begin{aligned} A_q(x) &= z_x d_x, \\ J_q &= \int_0^1 \min\{E_q(x), T_q(x), O_q(x), S_q(x), A_q(x)\} dx, \\ I_q &= \int_0^1 \min\left\{d_x, \frac{E_q(x)}{z_x}, \frac{T_q(x)}{z_x}, \frac{O_q(x)}{z_x}, \frac{S_q(x)}{z_x}\right\} dx. \end{aligned} \quad (\text{RMT13})$$

These are actual quantities of the original finite source path. The projections are continuous matrix functions of x . Explicitly $s_x = 2(q+1) - \text{rank}[U_x W_x]$. A rank superlevel set is a finite union of sets where a specified minor is nonzero, hence is Borel. This proves measurability of s_x and of every displayed integrand. Equation (RMT12) and compactness give a positive lower bound for $\mathcal{B}$ on this fixed path. Hence all denominators in I_q are nonzero, and the integrands are finite and bounded. This is a fixed-source statement and asserts no bound uniform in the tensor order.

All five preceding estimates bound the same $|\mathcal{B}'|$. Pointwise comparison followed by the triangle inequality for the integral therefore proves

$$\begin{aligned} |\mathcal{B}(1) - \mathcal{B}(0)| &\leq J_q, \\ J_q &\leq \min\left\{\mathcal{L}_q(D), \int_0^1 T_q(x) dx, \int_0^1 O_q(x) dx, \int_0^1 S_q(x) dx, \int_0^1 A_q(x) dx\right\}. \end{aligned} \quad (\text{RMT14})$$

where the earlier AW13 quantity is retained in full:

$$\mathcal{L}_q(D) = \log \frac{b_{q+2}}{b_q} + 2 \sum_{j=1}^{q-1} \log \frac{b_{2q+2-j}}{b_j} \leq (2q-1) \log \kappa. \quad (\text{RMT15})$$

Indeed integration of each c_j gives $\log b_j$, which proves the first entry of the minimum in (RMT14).

For $B > 0$ let

$$\mathcal{H}_a(B) = 2 \operatorname{arcosh}(e^{B/(2a)}), \quad \mathcal{H}'_a(B) = \frac{1}{a\sqrt{1 - e^{-B/a}}}, \quad \mathcal{H}_a^{-1}(z) = 2a \log \cosh(z/2) \quad (z > 0). \quad (\text{RMT16})$$

The derivative follows by differentiating the positive real inverse branch of $\cosh$; substitution proves both inverse composites. Applying this exact derivative to each of the five bounds on B' gives

$$|\mathcal{H}_a(\mathcal{B}(1)) - \mathcal{H}_a(\mathcal{B}(0))| \leq I_q \leq \int_0^1 dx \, dx = \log \kappa. \quad (\text{RMT17})$$

Thus the exact trace norm, supplied variance estimate, overlap and full-relative-spectrum information propagate into the nonlinear bound, not only into its linear consequence.

Writing $H_0 = \mathcal{H}_a(\mathcal{B}(0))$, the inverse in (RMT16), extended continuously by value zero at zero, and the additive estimate in (RMT14) give the simultaneous interval

$$\begin{aligned} \mathcal{B}(1) &\geq \max \left\{ 0, \mathcal{B}(0) - J_q, 2a \log \cosh \left(\frac{\max\{0, H_0 - I_q\}}{2} \right) \right\}, \\ \mathcal{B}(1) &\leq \min \left\{ \mathcal{B}(0) + J_q, 2a \log \cosh \left(\frac{H_0 + I_q}{2} \right) \right\}. \end{aligned} \quad (\text{RMT18})$$

The actual $\mathcal{B}(1)$ is strictly positive although the lower bound can equal zero. If $M_1 = cM_0$ with $c > 0$, all the c_j coincide, so $d_x, E_q, T_q, O_q, S_q, A_q, I_q, J_q$ vanish. Each original quotient determinant is multiplied by the same power q of $1 - x + xc$; the four signs then give $\mathcal{B}(1) = \mathcal{B}(0)$, as (RMT18) also states. This includes $q = 1$, for which the ordered sums above are empty and $a = 1$.

Every projection and metric in this argument uses the original J_N , B_N , coefficient basis and source measure. The arithmetic class map $\sigma_h^{\otimes k} \eta_{h,k} J_N$, its complete Taylor unit, and the original theta primitives of differences of minimum sections are unchanged. The proof refines numerical bounds on those exact maps; it introduces no identification between their source kernels or between distinct support labels. The earlier complete AW, SP and EP proofs, together with the explicit correspondences (RMT3)–(RMT7), provide the full input proofs used in this propagation.

3.4 Carrying the constructed signed residual into the endpoint

The complete supplied projector correspondence and its explicit finite approximation are proved in Section 4, (ISM14)–(ISM37). Their notation has the exact identities

$$X_x = C_x, \quad \mathcal{A}_x = U_x - W_x, \quad \Pi_N = P_N - Q_N.$$

For completeness retain the actual finite construction here. For each integer $L \geq 0$, put

$$\begin{aligned} B_x^\circ &= (1-x)I + xD, \quad \alpha_x = \frac{2(1-x) + x(b_1 + b_{2q+1})}{2}, \\ H_x^\circ &= I - B_x^\circ/\alpha_x, \quad C_x^{[L]} = \alpha_x^{-1} \sum_{r=0}^L (H_x^\circ)^r (D - I), \\ R_x^{[L]} &= (H_x^\circ)^{L+1} C_x, \quad E_x^{[L]} = R_x^{[L]} - \frac{\operatorname{Tr} R_x^{[L]}}{2q+1} I. \end{aligned} \quad (\text{RMT19})$$

The symbols $B_x^\circ, H_x^\circ, \alpha_x$ label operators and the scalar in this calculation; they leave every original relation matrix B_N , source metric M_x , and coordinate in (RMT1) in place. Multiplication of the finite geometric sum by $I - H_x^\circ$ gives $C_x - C_x^{[L]} = R_x^{[L]}$ exactly. Every factor is a real function of the same M_0 -self-adjoint D , hence is self-adjoint in $M_x = M_0 B_x^\circ$ as well. Since the trace of $U_x - W_x$ vanishes, (RMT7) becomes

$$\mathcal{B}'(x) = \text{Tr}(C_x^{[L]}(U_x - W_x)) + \text{Tr}(E_x^{[L]}(U_x - W_x)).$$

Apply Hilbert–Schmidt Cauchy–Schwarz in the original M_x metric to the last term and integrate. Define the specified real numbers

$$\begin{aligned} j_L &= \int_0^1 \text{Tr}(C_x^{[L]}(U_x - W_x)) dx, \\ e_L &= \int_0^1 \sqrt{\text{Tr}((E_x^{[L]})^2) \text{Tr}((U_x - W_x)^2)} dx, \\ \vartheta &= \frac{b_{2q+1} - b_1}{b_{2q+1} + b_1}, \quad K_D = \sum_{j=1}^{2q+1} \left(\frac{|b_j - 1|}{\min\{1, b_j\}} \right)^2. \end{aligned} \tag{RMT20}$$

These are J_L, e_L, ϑ, K_D of (ISM34)–(ISM37) with j_L, e_L used to avoid collision with J_q in (RMT13). The eigenvalues of H_x° have absolute value at most $\vartheta < 1$; those of C_x are the full $c_j(x)$. Thus $\text{Tr}((E_x^{[L]})^2) \leq \vartheta^{2L+2} K_D$, because subtracting the trace mean subtracts a nonnegative square from the full sum of squared residual eigenvalues. The two positive operators U_x, W_x dominate their range projections. Those ranges intersect in dimension at least one, so $\text{Tr}(U_x W_x) \geq 1$, by positivity of the trace product and the squared-length formula for the product of the range projections. Equation (RMT10) gives $\text{Tr}((U_x - W_x)^2) \leq 8q - 6$. The signed estimate and its computed residual are therefore

$$\begin{aligned} j_L - e_L &\leq \mathcal{B}(1) - \mathcal{B}(0) \leq j_L + e_L, \\ 0 \leq e_L &\leq \vartheta^{L+1} \sqrt{K_D(8q - 6)} \xrightarrow{L \rightarrow \infty} 0. \end{aligned} \tag{RMT21}$$

Every integral is a continuous finite matrix expression. This proves convergence for the fixed actual two source forms; no uniformity in k is assumed or inferred.

Let $B_0 = \mathcal{B}(0)$ and retain H_0 from (RMT18). Intersecting the proved intervals, for every $L \geq 0$, gives the propagated endpoint bounds

$$\begin{aligned} \mathcal{B}(1) &\geq \max \left\{ 0, B_0 - J_q, 2a \log \cosh \left(\frac{\max\{0, H_0 - I_q\}}{2} \right), B_0 + j_L - e_L \right\}, \\ \mathcal{B}(1) &\leq \min \left\{ B_0 + J_q, 2a \log \cosh \left(\frac{H_0 + I_q}{2} \right), B_0 + j_L + e_L \right\}. \end{aligned} \tag{RMT22}$$

Indeed each lower entry has separately been proved no larger than the actual endpoint, and each upper entry no smaller. Taking their maximum and minimum therefore preserves both inequalities and yields a nonempty interval containing that endpoint. A numerically integrated center must retain its own integration error before use; (RMT20)–(RMT22) state the exact integrals. The scalar source case has $H_x^\circ = E_x^{[L]} = 0$ and $j_L = e_L = 0$, so it also satisfies (RMT22) exactly.

4 Source enclosures and convergent signed projector control

This chapter incorporates the complete written derivations in the user-supplied continuation following merged PR29. The canonical secant and quotient-variation identities already occur in that PR. The four-projector pairing is the same pairing as AW and SP, with the exact coordinate correspondence proved below. The source sandwich, restriction-loss expression for its Hilbert–Schmidt size, and centered signed enclosure are retained in their full forms. We additionally construct the

approximating operator in that enclosure and prove an explicit convergent residual bound. The supplied Lean draft is preserved separately as uncompiled; the mathematical scope here is the written proof.

4.1 The unchanged polynomial and arithmetic source

Fix the actual full packet h , tensor order $k \geq 1$, and its monic cyclic polynomial $\chi = \chi_{h,k}$ of degree $q \geq 1$. Put $E = \mathbb{C}[S]/(\chi)$, $H = \mathcal{P}_{2q}$, and $n = 2q + 1$. All coordinates are the original ascending monomial coordinates. The original analytic source and observation are

$$\begin{aligned} g &= 2\xi, & \mathcal{M}F_h &= g/h, & \mathcal{V}P &= P(D_1 + \cdots + D_k)F_h^{\otimes k}, \\ J^{(k)}\mathcal{V}P &= \eta_{h,k}\pi_\chi P, & \eta_{h,k}[P] &= v_h^{\otimes k}P(A_h^{(k)})1, & v_h &= j_h(g/h), & A_h^{(k)} &= \sum_{i=1}^k A_{h,i}. \end{aligned} \quad (\text{ISM1})$$

Here A_h is multiplication by the original coordinate on $E_h = \mathbb{C}[s]/(h)$. Thus the domain of $\eta_{h,k}$ is E , its codomain is $E_h^{\otimes k}$, and its full unit is retained. The minimal polynomial definition of χ on the cyclic vector 1 proves that $[P] \mapsto P(A_h^{(k)})1$ is injective; multiplication by the invertible $v_h^{\otimes k}$ preserves injectivity.

For $N \in \{q-1, q, 2q-1, 2q\}$, let $I_N : \mathcal{P}_N \hookrightarrow H$ be coefficient inclusion, $B_N : \mathcal{P}_{N-q} \rightarrow H$ multiplication by χ followed by inclusion, and $C : E \rightarrow H$ the degree-less-than- q remainder lift in the retained quotient basis. The relation space is zero when $N = q-1$. Polynomial division by the monic χ proves the exact sequence

$$0 \longrightarrow \mathcal{P}_{N-q} \xrightarrow{\times\chi} \mathcal{P}_N \xrightarrow{\pi_\chi} E \longrightarrow 0. \quad (\text{ISM2})$$

Indeed division writes every polynomial uniquely as $\chi Q + r$, $\deg r < q$; if its class is zero, $r = 0$. This also proves injectivity of the relation map and every dimension used below.

Let M_0, M_1 be the actual positive definite Hermitian forms on this same H . In the arithmetic comparison these are the moments of $r_{1/4}^{*k}(y) dy$ and $w_h^{*k}(y) dy$ in $S = k/2 + iy$, with $w_h(y) = |(g/h)(1/2 + iy)|^2/(2\pi)$. The finite identities below also hold for any two positive forms on this unchanged presentation. The canonical section, its relation coefficient, and its positive quotient Gram are

$$\begin{aligned} Z_N(M) &= (B_N^* M B_N)^{-1} B_N^* M C, \\ \rho_N(M) &= C - B_N Z_N(M) : E \longrightarrow H, \\ G_N(M) &= \rho_N(M)^* M \rho_N(M), \quad V_N(M) = \det G_N(M) > 0. \end{aligned} \quad (\text{ISM3})$$

At $N = q-1$ the first map is the unique map to the zero space and $B_N Z_N = 0$; no preceding inverse is used. For every other degree, injectivity of B_N and positivity of M make its relation Gram positive definite. Direct substitution gives $B_N^* M \rho_N = 0$ and $\pi_\chi \rho_N = 1_E$. Hence ρ_N is injective, which proves positivity of G_N .

4.2 Both scaled secants, with the original boundary terms

Fix one admitted N , write $\rho_a = \rho_N(M_a)$, and put

$$\Delta = \rho_1 - \rho_0 = B_N(Z_N(M_0) - Z_N(M_1)). \quad (\text{ISM4})$$

For every coefficient map $Z : E \rightarrow \mathcal{P}_{N-q}$, the two cross terms in the Gram expansion vanish by $B_N^* M_a \rho_a = 0$, giving

$$(\rho_a + B_N Z)^* M_a (\rho_a + B_N Z) = G_N(M_a) + (B_N Z)^* M_a (B_N Z). \quad (\text{ISM5})$$

Taking $a = 0$, $B_N Z = \Delta$, and then $a = 1$, $B_N Z = -\Delta$, proves

$$\rho_1^* M_0 \rho_1 = G_0 + \Delta^* M_0 \Delta, \quad \rho_0^* M_1 \rho_0 = G_1 + \Delta^* M_1 \Delta. \quad (\text{ISM6})$$

Here $G_a = G_N(M_a)$. Subtracting the displayed matrices without deleting either boundary term yields, for real a, b ,

$$\begin{aligned} G_1 - aG_0 &= \rho_1^*(M_1 - aM_0)\rho_1 + a\Delta^* M_0 \Delta, \\ bG_0 - G_1 &= \rho_0^*(bM_0 - M_1)\rho_0 + \Delta^* M_1 \Delta. \end{aligned} \quad (\text{ISM7})$$

In particular both boundary terms have the positive signs shown. For $0 < a \leq b$ and $aM_0 \preceq M_1 \preceq bM_0$, congruence of positive matrices and (ISM7) give

$$aG_N(M_0) \preceq G_N(M_1) \preceq bG_N(M_0) \quad \text{at every one of the four degrees.} \quad (\text{ISM8})$$

Such finite endpoints always exist: they may be taken to be the smallest and largest eigenvalues of $M_0^{-1}M_1$, which is positive self-adjoint in the original M_0 metric.

Conjugating (ISM8) by $G_N(M_0)^{-1/2}$ places every eigenvalue of the conjugated positive matrix in $[a, b]$. The determinant is its product and logarithm is the sum of the real logarithms, so

$$q \log a \leq \log \frac{V_N(M_1)}{V_N(M_0)} \leq q \log b. \quad (\text{ISM9})$$

Set

$$\mathcal{B}(M) = \log \frac{V_{q-1}(M)V_q(M)}{V_{2q-1}(M)V_{2q}(M)}, \quad \mathcal{C} = \mathcal{B}(M_1) - \mathcal{B}(M_0). \quad (\text{ISM10})$$

There are exactly two positive and two negative terms in $\mathcal{C}$. Applying the upper endpoint in (ISM9) to the positive terms and the lower endpoint to the negative ones, and then reversing these choices, proves

$$|\mathcal{C}| \leq 2q \log(b/a). \quad (\text{ISM11})$$

Consequently a certified common source enclosure $(1 - \varepsilon)\widehat{M} \preceq M \preceq (1 + \varepsilon)\widehat{M}$, $0 \leq \varepsilon < 1$, gives

$$|\mathcal{B}(M) - \mathcal{B}(\widehat{M})| \leq 2q \log \frac{1 + \varepsilon}{1 - \varepsilon}. \quad (\text{ISM12})$$

These are the supplied source-error estimates in their original forms. At these same use sites the stronger previously proved AW/SP estimate must also be retained. For all ordered eigenvalues $\beta_1 \leq \dots \leq \beta_{2q+1}$ of $D = M_0^{-1}M_1$, define its full existing expression

$$\mathcal{L}_q(D) = \log \frac{\beta_{q+2}}{\beta_q} + 2 \sum_{j=1}^{q-1} \log \frac{\beta_{2q+2-j}}{\beta_j}.$$

The sum is empty at $q = 1$. AW's two nested-flag trace estimates, with the exact operator identification (ISM16)–(ISM19), give

$$|\mathcal{C}| \leq \mathcal{L}_q(D) \leq (2q - 1) \log(b/a).$$

For the last inequality every displayed eigenvalue ratio is at most b/a , and its total coefficient is $1 + 2(q - 1) = 2q - 1$. Thus the same source enclosure in (ISM12) also gives the stronger $(2q - 1) \log((1 + \varepsilon)/(1 - \varepsilon))$ bound.

For the actual moment paths of RMT, let J_q be its integral of the pointwise minimum of the full-spectrum, exact trace-norm, overlap, centered Hilbert–Schmidt, and angle bounds. Each bounds the identical derivative (ISM19); the Hilbert–Schmidt member is (ISM26). Taking their pointwise minimum before integrating gives the recursively refined source-error statement

$$\begin{aligned} |\mathcal{C}| &\leq \min\{J_q, \mathcal{L}_q(D), (2q - 1) \log(b/a)\}, \\ \mathcal{C} &\in [J_L - \epsilon_L, J_L + \epsilon_L] \cap [-J_q, J_q] \quad (L \geq 0). \end{aligned}$$

Here J_L, ϵ_L are the fully constructed signed center and residual in (ISM31)–(ISM37). Their interval therefore retains its original orientation in this earlier source-error calculation. For generic positive coefficient forms, the full-spectrum, trace-norm, overlap, Hilbert–Schmidt and signed bounds still apply, but the moment-specific angle improvement and its strict moment-positivity proof are used only on the original moment paths. The exact primitive and coefficient support are established in (ISM38)–(ISM43). Scaling a source by $c > 0$ leaves Z_N, ρ_N unchanged and multiplies G_N, V_N by c, c^q , respectively, directly by (ISM3). Thus its four $q \log c$ terms cancel in (ISM10); this proves the scalar invariance without altering any source form.

4.3 The canonical projector is exactly the original difference

Use $x \in [0, 1]$ for the comparison path, retaining the frequency y and all other deformation parameters. Write

$$M_x = (1 - x)M_0 + xM_1, \quad \dot{M} = M_1 - M_0, \quad X_x = M_x^{-1}\dot{M}. \quad (\text{ISM13})$$

Since $M_x X_x = \dot{M} = \dot{M}^*$, this is self-adjoint in the original M_x inner product. Define

$$\begin{aligned} P_N &= I_N (I_N^* M_x I_N)^{-1} I_N^* M_x, \\ Q_N &= B_N (B_N^* M_x B_N)^{-1} B_N^* M_x, \quad Q_{q-1} = 0, \\ \Pi_N &= \rho_N G_N^{-1} \rho_N^* M_x. \end{aligned} \quad (\text{ISM14})$$

Each formula squares to itself and has M_x -self-adjoint adjoint law. The ranges of the first two are $\mathcal{P}_N$ and $B_N \mathcal{P}_{N-q}$; the range of the third is $\rho_N E$. By (ISM2)–(ISM3),

$$\mathcal{P}_N = \rho_N E \perp_{M_x} B_N \mathcal{P}_{N-q}. \quad (\text{ISM15})$$

To prove the equality, subtract $\rho_N \pi_\chi p$ from any $p \in \mathcal{P}_N$; (ISM2) puts the result in the relation range, and (ISM3) proves orthogonality. Therefore the original P_N, Q_N commute with product Q_N , and their orthogonal range decomposition proves the exact operator identity

$$\Pi_N = P_N - Q_N, \quad \text{Tr } \Pi_N = q. \quad (\text{ISM16})$$

The inverse of the range parametrization $\rho_N : E \rightarrow \text{ran } \Pi_N$ is the literal π_χ restricted to that range; both composites are identities by (ISM3). Thus (ISM16) identifies actual maps and their domains, not only their traces.

Define

$$\begin{aligned} \mathcal{A}_x &= \Pi_{q-1} + \Pi_q - \Pi_{2q-1} - \Pi_{2q}, \\ U_x &= Q_{2q-1} + Q_{2q} - Q_q, \\ W_x &= (P_{2q-1} - P_{q-1}) + (P_{2q} - P_q). \end{aligned} \quad (\text{ISM17})$$

Substitution of (ISM16), with $Q_{q-1} = 0$, gives $\mathcal{A}_x = U_x - W_x$ and $\text{Tr } \mathcal{A}_x = 0$. These U_x, W_x are AW's U, W and SP's $\mathcal{R}, \mathcal{P}$, respectively. The source relation matrix is related to AW's unpadded matrix by $B_N = I_N B_N^{\text{AW}}$, so (ISM14) reproduces its actual projections with no sign or metric change.

Differentiating $\pi_\chi \rho_N = 1$ shows that ρ'_N has image in the original relation space. More explicitly, differentiation of $B_N^* M_x \rho_N = 0$ gives

$$\rho'_N = -B_N (B_N^* M_x B_N)^{-1} B_N^* \dot{M} \rho_N, \quad G'_N = \rho_N^* \dot{M} \rho_N. \quad (\text{ISM18})$$

The second identity follows because the two cross terms containing ρ'_N vanish by the same orthogonality. All these matrices are smooth on the closed interval, since their positive Gram inverses exist there. Jacobi differentiation and cyclicity of finite trace prove

$$\frac{d}{dx} \log V_N = \text{Tr}(X_x \Pi_N), \quad \mathcal{B}'(x) = \text{Tr}(X_x \mathcal{A}_x), \quad \mathcal{C} = \int_0^1 \text{Tr}(X_x \mathcal{A}_x) dx. \quad (\text{ISM19})$$

For the original row $v(y) = (1, S, \dots, S^{2q})$, $\Pi_N M_x^{-1} = \rho_N G_N^{-1} \rho_N^*$. Thus the signed density in PR29 is exactly

$$\kappa_x(y) = v(y) \mathcal{A}_x M_x^{-1} v(y)^*, \quad \int \kappa_x d\mu_x = \text{Tr } \mathcal{A}_x = 0. \quad (\text{ISM20})$$

The first equality keeps the inverse metric that is required by the original coordinate kernel. Integrating against the original signed measure $\mu_1 - \mu_0$ gives precisely (ISM19).

4.4 Restriction losses and the full cross trace

Fix $i \leq j$ among admitted degrees. The map $\rho_i - \rho_j : E \rightarrow \mathcal{P}_j$ has zero remainder, so (ISM2) makes it an original degree- j relation. Orthogonality of ρ_j therefore gives the cross-Grams

$$\rho_j^* M_x \rho_i = G_j, \quad \rho_i^* M_x \rho_j = G_j, \quad G_i - G_j = (\rho_i - \rho_j)^* M_x (\rho_i - \rho_j). \quad (\text{ISM21})$$

The exact reverse restriction on E is $T_{i,j} = G_i^{-1} G_j$. It is positive self-adjoint in G_i , with spectrum in $(0, 1]$, by the last identity. Cyclic trace and the first two identities give

$$\begin{aligned} \text{Tr}(\Pi_i \Pi_j) &= \text{Tr}(G_i^{-1} (\rho_i^* M_x \rho_j) G_j^{-1} (\rho_j^* M_x \rho_i)) \\ &= \text{Tr}(G_i^{-1} G_j), \\ \text{Tr}((\Pi_i - \Pi_j)^2) &= 2 \text{Tr}(1 - T_{i,j}). \end{aligned} \quad (\text{ISM22})$$

All traces are finite traces of the original coordinate operators. In particular their spectra are not declared to be arithmetic invariant summands.

Set

$$\begin{aligned} A_0 &= \Pi_{q-1} - \Pi_{2q-1}, \quad A_1 = \Pi_q - \Pi_{2q}, \\ \ell_0 &= \text{Tr}(1 - T_{q-1, 2q-1}), \quad \ell_1 = \text{Tr}(1 - T_{q, 2q}), \\ Z_x &= \text{Tr}(\mathcal{A}_x^2) = 2\ell_0 + 2\ell_1 + 2\text{Tr}(A_0 A_1). \end{aligned} \quad (\text{ISM23})$$

This retains the cross term with its actual sign. On the full algebra $\text{End}(H)$, the map $L \mapsto M_x^{1/2} L M_x^{-1/2}$ is an algebra isomorphism with inverse $K \mapsto M_x^{-1/2} K M_x^{1/2}$. It sends the M_x -self-adjoint operators to Hermitian operators, and cyclicity preserves traces of every product. This proves $Z_x \geq 0$ and, from $\|A_0 + A_1\|_{\text{HS}}^2 \leq 2\|A_0\|_{\text{HS}}^2 + 2\|A_1\|_{\text{HS}}^2$, proves

$$0 \leq Z_x \leq 4(\ell_0 + \ell_1). \quad (\text{ISM24})$$

Here and below the Hilbert–Schmidt norm is computed after this exact metric isomorphism; no new source metric is used.

Put

$$c_x = \frac{\text{Tr } X_x}{n}, \quad \Sigma_x = \text{Tr}(X_x^2) - \frac{(\text{Tr } X_x)^2}{n}. \quad (\text{ISM25})$$

The trace is real by self-adjointness. Expanding $\text{Tr}((X_x - c_x I)^2)$ proves its equality with Σ_x , hence its nonnegativity. Trace zero of $\mathcal{A}_x$, and Cauchy–Schwarz for the two Hermitian images in the preceding isomorphism, give

$$|\mathcal{B}'(x)| \leq S_q(x) := \sqrt{\Sigma_x Z_x} \leq 2\sqrt{\Sigma_x(\ell_0 + \ell_1)}. \quad (\text{ISM26})$$

This bound can be combined pointwise with the existing full-spectrum, exact trace-norm, overlap, and angle bounds on the same $\mathcal{B}'$. It does not replace any smaller available bound.

4.5 A specified signed approximation and its exact residual

For an M_x -self-adjoint, continuous operator Y_x define $E_x = X_x - c_x I - Y_x$. It is self-adjoint too. Expanding the trace and applying the same Cauchy–Schwarz inequality proves the two signed inequalities

$$\text{Tr}(Y_x \mathcal{A}_x) - \sqrt{\text{Tr}(E_x^2) Z_x} \leq \mathcal{B}'(x) \leq \text{Tr}(Y_x \mathcal{A}_x) + \sqrt{\text{Tr}(E_x^2) Z_x}. \quad (\text{ISM27})$$

Every term is real. Integration is legitimate by continuity on the compact interval. We now construct such a Y_x , with its residual controlled from the actual two endpoint forms.

Let $D = M_0^{-1}M_1$, with ordered positive eigenvalues $b_1 \leq \dots \leq b_n$, counted with multiplicity. For this subsection write $B_x^\circ = (1-x)I + xD$, to distinguish it from every original relation matrix. Since $M_x = M_0 B_x^\circ$, direct inversion gives

$$X_x = (B_x^\circ)^{-1}(D - I), \quad \lambda_j(x) = 1 - x + xb_j, \quad \gamma_j(x) = \frac{b_j - 1}{1 - x + xb_j}. \quad (\text{ISM28})$$

Take an M_0 -orthogonal eigenbasis of D . It is also M_x -orthogonal, since on its j -th line M_x multiplies M_0 by $\lambda_j(x) > 0$. Therefore (ISM28) gives the full spectrum of X_x in the original varying metric.

Define the positive scalar and commuting operator

$$a_x = \frac{\lambda_1(x) + \lambda_n(x)}{2}, \quad H_x = I - B_x^\circ/a_x. \quad (\text{ISM29})$$

Their use does not rescale M_x , its moments, or its quotient Grams. The eigenvalues of H_x are $\zeta_j(x) = 1 - \lambda_j(x)/a_x$, and

$$\|H_x\|_{M_x} = \frac{x(b_n - b_1)}{2(1-x) + x(b_1 + b_n)} \leq \vartheta := \frac{b_n - b_1}{b_n + b_1} < 1. \quad (\text{ISM30})$$

Indeed the extrema of $1 - \lambda/a_x$ are equal and opposite at the two endpoints. The displayed scalar function of x has derivative $2(b_n - b_1)/(2(1-x) + x(b_1 + b_n))^2 \geq 0$. This proves the uniform bound, including $x = 0$.

For every integer $L \geq 0$, form the finite matrix polynomial and its centered version

$$\begin{aligned} X_x^{[L]} &= a_x^{-1} \sum_{r=0}^L H_x^r (D - I), \\ Y_x^{[L]} &= X_x^{[L]} - \frac{\text{Tr } X_x^{[L]}}{n} I. \end{aligned} \quad (\text{ISM31})$$

All factors commute with D and are real functions of its self-adjoint operator. They are consequently self-adjoint in both M_0 and M_x . The exact finite geometric-series identity $(I - H_x) \sum_{r=0}^L H_x^r = I - H_x^{L+1}$, together with $B_x^\circ = a_x(I - H_x)$, proves

$$\begin{aligned} R_x^{[L]} &:= X_x - X_x^{[L]} = H_x^{L+1} X_x, \\ E_x^{[L]} &:= X_x - c_x I - Y_x^{[L]} = R_x^{[L]} - \frac{\text{Tr } R_x^{[L]}}{n} I. \end{aligned} \quad (\text{ISM32})$$

No infinite-series interchange is used in this identity. In the original eigenbasis the residual eigenvalues are $\delta_j(x) = \zeta_j(x)^{L+1} \gamma_j(x)$. Thus its exact centered squared size is

$$\text{Tr}((E_x^{[L]})^2) = \sum_{j=1}^n \delta_j(x)^2 - \frac{(\sum_{j=1}^n \delta_j(x))^2}{n}. \quad (\text{ISM33})$$

Both sides are nonnegative, and all multiplicities and signs of the residual eigenvalues are retained. The signed center is $\text{Tr}(Y_x^{[L]} \mathcal{A}_x) = \text{Tr}(X_x^{[L]} \mathcal{A}_x)$ by $\text{Tr } \mathcal{A}_x = 0$. Equations (ISM27), (ISM31)–(ISM33) therefore provide a finite constructed signed center and its exact error.

There is also a uniform elementary error bound for this fixed original pair of forms. Define

$$K_D = \sum_{j=1}^n \left(\frac{|b_j - 1|}{\min\{1, b_j\}} \right)^2. \quad (\text{ISM34})$$

The denominator of $\gamma_j(x)$ is at least $\min\{1, b_j\}$, so (ISM30)–(ISM33) imply $\text{Tr}((E_x^{[L]})^2) \leq \vartheta^{2L+2} K_D$. The nested relation and source flags in (ISM17) give both U_x, W_x eigenvalues $0^{[q]}, 1^{[2]}, 2^{[q-1]}$: the first relation line contributes one, its next $q-1$ increments contribute two, and the final increment contributes one; the source increments give the same list. In particular both have rank $q+1$, and each dominates its range projection. Their ranges in dimension $2q+1$ intersect in dimension at least one. The trace product of the two range projections is at least that intersection dimension, by summing squared projected lengths in an orthonormal basis. For positive operators, increasing a factor cannot decrease the product trace, since $\text{Tr}(T^{1/2}(S' - S)T^{1/2}) \geq 0$. It follows that $\text{Tr}(U_x W_x) \geq 1$. Using their complete spectra, we obtain

$$Z_x = 8q - 4 - 2\text{Tr}(U_x W_x) \leq 8q - 6. \quad (\text{ISM35})$$

This proof includes $q = 1$, where the multiplicity $q - 1$ is zero.

Set

$$\begin{aligned} J_L &= \int_0^1 \text{Tr}(X_x^{[L]} \mathcal{A}_x) dx, \\ \epsilon_L &= \int_0^1 \sqrt{\text{Tr}((E_x^{[L]})^2) Z_x} dx. \end{aligned} \quad (\text{ISM36})$$

Every integrand is a specified continuous finite matrix expression on the original source path. Integrating (ISM27) with the constructed (ISM31) and the established uniform bound gives

$$J_L - \epsilon_L \leq \mathcal{C} \leq J_L + \epsilon_L, \quad 0 \leq \epsilon_L \leq \vartheta^{L+1} \sqrt{K_D(8q-6)} \xrightarrow{L \rightarrow \infty} 0. \quad (\text{ISM37})$$

Thus the supplied signed enclosure now has an explicit convergent residual on every fixed original finite source. It retains the arithmetic correlation in its center. The evaluation of the center integral still uses the specified arithmetic entries and a controlled integration method; this calculation supplies no uniform estimate in k . If $b_n = b_1$, then $D = b_1 I$, $H_x = 0$, the residual and $\mathcal{C}$ vanish exactly, and every formula above remains valid.

4.6 The original primitive and the residual at a proper support

We finally prove the source comparison underlying all these metric changes. In the original one-factor complex $V \xrightarrow{\Theta} \mathcal{B}$, retain $h(D)F_h = \Theta\phi_*$, with the original D action commuting with Θ . For $v \in E$, the polynomial $(\rho_1 - \rho_0)v = \chi Q_v$ has the coefficient $Q_v = (Z_N(M_0) - Z_N(M_1))v$ by (ISM4). In the multivariate polynomial ring put

$$f_v(s_1, \dots, s_k) = \chi(s_1 + \dots + s_k) Q_v(s_1 + \dots + s_k). \quad (\text{ISM38})$$

Its image in $E_h^{\otimes k}$ is zero, because $\chi(A_h^{(k)})1 = 0$. Successive monic division in $s_1, \dots, s_k$ constructs specific polynomials $Q_{v,i}$ as follows. Let $f_{v,0} = f_v$ and write $f_{v,i-1} = h(s_i)Q_{v,i} + f_{v,i}$, where $\deg_{s_i} f_{v,i} < \deg h$. Earlier remainder degree bounds are preserved because $h(s_i)$ uses only s_i . The final remainder $f_{v,k}$ has degree below $\deg h$ in every variable and zero class in the quotient, so it is zero by the tensor monomial basis. Telescoping proves

$$f_v = \sum_{i=1}^k h(s_i) Q_{v,i}. \quad (\text{ISM39})$$

This constructs a primitive using precisely the original relations, without replacing them by a larger observation kernel.

In the tensor complex $\mathbf{C}^\bullet = [V \xrightarrow{\Theta} \mathcal{B}]^{\otimes k}$ define

$$K_v = \sum_{i=1}^k (-1)^{i-1} Q_{v,i}(D_1, \dots, D_k) (F_h^{\otimes(i-1)} \otimes \phi_* \otimes F_h^{\otimes(k-i)}) \in \mathbf{C}^{k-1}. \quad (\text{ISM40})$$

The displayed source tensor has exactly $i-1$ preceding factors of degree one. Its differential therefore has sign $(-1)^{i-1}$; multiplying by the sign in (ISM40) gives $+1$. The source operators commute with the differential. Applying $h(D_i)F_h = \Theta\phi_*$ to (ISM39) now proves

$$dK_v = \mathcal{V}((\rho_1 - \rho_0)v). \quad (\text{ISM41})$$

Consequently the two canonical sections have the same class in the original full-source quotient; their common arithmetic observation is exactly $\eta_{h,k}v$ by (ISM1). Composing with the original full-source quotient $q^{(k)}$ gives the already constructed map $\sigma_h^{\otimes k} \eta_{h,k} : E \hookrightarrow Q^{\otimes k}$. Here no map was made natural merely by changing its metric; the actual primitive (ISM40) is the comparison.

To express the full two-leg source without deleting its diagonal, retain the labelled Fourier copy $\widehat{V}$, with the original isomorphism $\mathcal{F} : V \rightarrow \widehat{V}$, and the complex $\mathbf{T}^\bullet = [V \oplus \widehat{V} \rightarrow \mathcal{B}]$, whose differential is $(\phi, \widehat{\psi}) \mapsto \Theta(\phi - \mathcal{F}^{-1}\widehat{\psi})$. There are exact cochain maps from the one-leg complex to this complex with degree-one map identity and degree-zero maps

$$j_+(v) = (v, 0), \quad j_-(v) = (0, -\mathcal{F}v).$$

Indeed either differential equals Θv . The reverse cochain map has degree-one map identity and degree-zero map $p(\phi, \widehat{\psi}) = \phi - \mathcal{F}^{-1}\widehat{\psi}$. Thus $pj_+ = pj_- = 1$, whereas $(j_+ - j_-)(v) = (v, \mathcal{F}v)$ is the full degree-zero diagonal kernel. These formulas prove both composites and retain that kernel explicitly. Tensoring specified choices of these degree-zero maps with their degree-one identities gives cochain maps, since the displayed differential identities preserve each original tensor sign. Apply one such map to (ISM40) to obtain $\widetilde{K}_v \in (\mathbf{T}^{\otimes k})^{k-1}$. Its top differential is still (ISM41).

For precision at a proper declared support $\lambda = (W, S; A)$, let $\mathcal{J}$ be the specified finite carrier of coefficient positions with $A \subseteq \mathcal{J}$. Put $\mathbf{C}_{\text{full}}^\bullet = (\mathbf{T}^{\otimes k})^{\oplus \mathcal{J}}$, with the original differential on each coordinate. Insert the active A -coordinates by literal zero insertion into this fixed full carrier. Let $\mathbf{C}_\lambda^\bullet \subseteq \mathbf{C}_{\text{full}}^\bullet$ be its declared subcomplex, retaining its independent outer label and coefficient mask. The same source primitive is taken in the indicated coefficient copy. When the compared top source representatives lie in $\mathbf{C}_\lambda^k$, their difference has the exact residual

$$[d\widetilde{K}_v]_\lambda \in \frac{d\mathbf{C}_{\text{full}}^{k-1} \cap \mathbf{C}_\lambda^k}{d\mathbf{C}_\lambda^{k-1}}. \quad (\text{ISM42})$$

This quotient is the kernel of the natural map $\mathbf{C}_\lambda^k/d\mathbf{C}_\lambda^{k-1} \rightarrow \mathbf{C}_{\text{full}}^k/d\mathbf{C}_{\text{full}}^{k-1}$: a class maps to zero exactly when its representative is a full-source boundary. The inclusion of the displayed numerator therefore induces the kernel isomorphism, with inverse obtained by using the same representative. These two maps are well-defined since differences in $d\mathbf{C}_\lambda^{k-1}$ are precisely the equivalence relation on both sides. Thus (ISM41) kills the class at this support exactly when $d\widetilde{K}_v \in d\mathbf{C}_\lambda^{k-1}$. Membership $\widetilde{K}_v \in \mathbf{C}_\lambda^{k-1}$ is one explicit sufficient witness, not an assumption made for arbitrary W .

First specialize to $k = 1$ and the single coefficient carrier $\mathcal{J} = \{i_0\}$, with active face $A = \{i_0\}$. The coordinate projection onto i_0 and its inverse coordinate insertion identify this one-coordinate carrier with the original one-factor complex. In these specified coordinates put $\mathcal{B}_\lambda = \mathbf{C}_\lambda^1 \subseteq \mathcal{B}$. Suppose the admitted source image under p is $W \subseteq V$, so its boundary range is exactly ΘW , and retain $\Theta V \subseteq \mathcal{B}_\lambda$. The precise kernel map in this single coordinate is

$$V/W \longrightarrow \mathcal{B}_\lambda/\Theta W, \quad [v] \longmapsto [\Theta v]_W. \quad (\text{ISM43})$$

It is injective because Θ is injective: $\Theta v = \Theta w$, $w \in W$, forces $v = w$. Its range is the kernel of the map to the full quotient $\mathcal{B}/\Theta V$, since being in that kernel means being represented by Θv . In this case (ISM42) is exactly the class of the original one-factor primitive in V/W .

For an arbitrary admitted face $A \subseteq \mathcal{J}$, restore the original coordinate injections $\iota_a : \mathcal{B} \rightarrow \mathcal{B}^{\oplus \mathcal{J}}$ and projections $\text{pr}_a : \mathcal{B}^{\oplus \mathcal{J}} \rightarrow \mathcal{B}$. They satisfy $\text{pr}_b \iota_a = \delta_{ab}1$ and $\sum_{a \in A} \iota_a \text{pr}_a = 1$ on the A -supported subspace. Let the degree-one source at this face be $\mathcal{B}_\lambda^{\oplus A}$, inserted in the fixed carrier

by these ι_a , and retain the degree-zero source image $W^{\oplus A}$ and boundary $(\Theta W)^{\oplus A}$. Its kernel map is

$$(V/W)^{\oplus A} \longrightarrow \mathcal{B}_\lambda^{\oplus A}/(\Theta W)^{\oplus A}, \quad ([v_a])_{a \in A} \longmapsto \left[\sum_{a \in A} \iota_a \Theta v_a \right].$$

Replacing v_a by $v_a + w_a$, $w_a \in W$, changes the displayed sum by the specified boundary, so the map is well-defined. If its image is zero, projection by each pr_a gives $\Theta v_a \in \Theta W$, whence $v_a \in W$ by injectivity of Θ . Thus the map is injective. A representative in its codomain maps to zero in $(\mathcal{B}/\Theta V)^{\oplus \mathcal{S}}$ exactly when each active coordinate is Θv_a for some $v_a \in V$. Its inverse on that kernel takes these unique preimages modulo W . Changing a representative by an element of $(\Theta W)^{\oplus A}$ changes each preimage by an element of W . The coordinate identities above then prove both inverse composites. This is the exact $(V/W)^{\oplus A}$ residual, with its full ambient carrier specified.

For $A \subseteq A'$, zero insertion sends the tuple $([v_a])_{a \in A}$ to the same tuple extended by zero; both orders of this insertion and the kernel map give $\sum_{a \in A} \iota_a \Theta v_a$. A signed coefficient permutation sends that sum to $\sum_{a \in A} \epsilon_a \iota_{\sigma(a)} \Theta v_a = \sum_{a \in A} \iota_{\sigma(a)} \Theta(\epsilon_a v_a)$, which proves the same commuting identity with every sign retained. When $A = \emptyset$, the two displayed direct sums and their unique map are zero vector spaces and a zero map at the retained outer label $(W, S; \emptyset)$; the coordinate calculation does not identify that label with external absence. This proves the typed relation to the existing mixed-support quotients while retaining every proper-source residual.

All source identities, projector equalities, scaled comparison signs, and signed estimates above use the same original finite polynomial presentation. Their full primitive and proper support kernel have now been exhibited, so backward and forward propagation can replace weaker metric bounds at their earlier use sites without erasing the older sources or claiming that the arbitrary proper-support residual vanishes.